

Problem Set 3 (two pages)

Statistics 24510-30040 (W25)

Due 9 am Tuesday, January 28.

Requirements

- Put the assignment number, course number (24510 or 30040) and your name at the top of the first page.
- Scan and save your work as a letter-size pdf file named "Lastname-Firstname-hw3.pdf" (abbreviations allowed for long names).
- Upload your pdf to Gradescope under Pset3. Be sure to mark the pages corresponding to each problem during submission.
- Show your work. Provide detailed derivations and arguments in order to receive credit.
- You may discuss methods with others, but you should devise and write your answers by yourself completely and independently.

Problem assignments (Related sections: Rice 4.3, 4.4; Casella and Berger 4.1-4.3.)

1. (t and χ^2 distributions)

Suppose $Z_i, i = 1, 2$ are independent, $Z_i \sim N(0, 1)$.

- (a) i. Show that $\frac{Z_1}{|Z_2|} \sim t_1$ (also called Cauchy distribution).
ii. Based on your result in part (i), derive the distribution of $\frac{Z_1}{Z_2}$.
- (b) Show that $\mathbb{P}(Z_i^2 \leq t) = 2\mathbb{P}(Z_i \leq \sqrt{t}) - 1$.

2. (Bivariate normal exercises)

An early criminologist believed he could summarize human characteristics for purposes of identification from a few simple measurements. Let

X = height, Y = length of left arm, W = width of skull, and Z = length of ear. Measurements are in centimeters.

Suppose (for a hypothetical population) (X, Y) and (W, Z) are bivariate normal vectors,

$$\begin{bmatrix} X \\ Y \end{bmatrix} \sim N \left(\begin{bmatrix} 140 \\ 70 \end{bmatrix}, \begin{bmatrix} 100 & 25 \\ 25 & 25 \end{bmatrix} \right), \quad \begin{bmatrix} W \\ Z \end{bmatrix} \sim N \left(\begin{bmatrix} 15 \\ 5 \end{bmatrix}, \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \right)$$

Suppose that (X, Y) and (W, Z) are independent. The criminologist proposed two summary measures,

$$M_1 = X + Y + W + Z, \quad M_2 = X - 2Y + W - 3Z$$

- (a) What are the expected averages of the two measures M_1 and M_2 ?
(b) What are the variances of the two measures M_1 and M_2 ?
(c) Find the bivariate normal distribution of (M_1, M_2) , in terms of the mean and covariance matrices.
(d) Are the two measures M_1 and M_2 very correlated? (Support your answer quantitatively.)
(e) What percentage of individuals in this population would have the measure M_1 greater than 250?

3. (Correlation of bivariate normal)

Suppose (X, Y) are of bivariate normal distribution with

expectations $(\mu_X, \mu_Y) = (3, 1)$, variance $(\sigma_X^2, \sigma_Y^2) = (9, 16)$, and correlation ρ .

Let $W_a = 5 + aX + Y$ and $V = 7 + X + 2Y$.

- (a) For $\rho = 1/3$, find $a \in \mathbb{R}$ such that W_a and V are independent.
- (b) Are there $\rho^* \in (-1, 1)$ such that no $a \in \mathbb{R}$ can make W_a and V independent?
If yes, find all such ρ^* . If no, explain why not (quantitatively).
- (c) Fix $a = 1$. Are there values of $\rho \in (-1, 1)$ such that W_a and V are independent?
If yes, find all such ρ . If no, explain why not.
- (d) Is there $a^* \in \mathbb{R}$ such that no $\rho \in (-1, 1)$ can make W_{a^*} and V independent?
If yes, find all such a^* . If no, explain why not.

4. (*Bivariate normal application*)

Suppose the natural logarithms of the heights of mothers and daughters (in centimeters) from a region follow a bivariate normal distribution with the mean for the mother of $\log_e(160)$, the mean for the daughters is $\log_e(165)$, the standard deviations for both distributions is 0.05 and the correlation is 0.5.

- (a) Find the probability that a mother is taller than her daughter.
- (b) Find the probability that a mother is at least 90% as tall as her daughter.
- (c) Give the joint density for the logarithms of the mother's and daughter's heights when heights is measured in inches (1 inch = 2.54 cm).
- (d) Find the density of the ratio of a mother's height to the daughter's height.

5. (*Geometric property of standardized bivariate normal*)

Let (U, V) be a standardized bivariate normal random vector, with $\mathbb{E}(U) = \mathbb{E}(V) = 0$, $\text{Var}(U) = \text{Var}(V) = 1$. Assume $\text{corr}(U, V) = \rho \in [0, 1)$.

- (a) Show $\mathbb{E}(UV) = \rho$ by solving an appropriate double integral of the joint density function of (U, V) .
- (b) Prove that on the (u, v) plane, the contours or the level curves of the density $\{(u, v) : f(u, v) = \text{constant}\}$ are ellipses, and $u = v$ is the direction of the major axis.
- (c) Find the (aspect) ratio of the lengths of major axis and minor axis of the elliptical contours.
- (d) How strongly correlated U and V has to be in order for the aspect ratio in (c) ≥ 2 ?

6. (*Normal sample mean properties*)

Suppose $X_1, \dots, X_n$ are *i.i.d.* $\sim N(\mu, 1)$. $\bar{X} = \sum_{i=1}^n X_i/n$.

- (a) Derive $\text{Cov}(X_1, \bar{X})$ and $\text{corr}(X_1, \bar{X})$.
- (b) $\begin{bmatrix} X_1 \\ \bar{X} \end{bmatrix}$ is of bivariate normal distribution. Find its mean vector and covariance matrix.
- (c) Find the conditional mean $\mathbb{E}(X_1 | \bar{X} = \bar{x})$ and the conditional subsample mean $\mathbb{E}(\frac{X_1 + X_2 + \dots + X_n}{n} | \bar{X} = \bar{x})$.
- (d) Find the conditional variance $\text{Var}(X_1 | \bar{X} = \bar{x})$.